

XIAOYING LIAO

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Research Interests: AI for Biology; Computational Biology; Single-Cell & Spatial Genomics

INTRODUCTION

I am a researcher passionate about exploring the unknown and taking on challenging scientific questions. My research lies at the intersection of AI, computational biology, and experimental biology, integrating computational modeling with single-cell and spatial genomics to understand complex biological systems and uncover new biological insights.

EDUCATION

Johns Hopkins University (JHU)

Aug. 2024 – May 2025

Master of Science in Engineering (M.S.E.) in Biomedical Engineering

Baltimore, USA

- Supervised by *Professor Carlo Iomini*, **GPA:** 4.00/4.00
- Courses:** Cellular Engineering, Tissue Engineering, Cell & Tissue Engineering Lab, Advanced Topics in Regenerative and Immune Engineering, Human Genetics, Systems Pharmacology and Personalized Medicine

University of Sydney (USYD)

Feb. 2020 – Dec. 2023

Bachelor of Science (Honours) in Statistics

Sydney, Australia

- Supervised by *Professor Jean Yang*, **GPA:** 3.90/4.00, **Rank:** 1/200
- Courses:** Probability and Estimation Theory, Statistical Inference, Stochastic Process, Advanced Statistical Modelling, Statistical Machine Learning, Life and Evolution

PUBLICATIONS

Python Song, **Xiaoying Liao**. *ProactiveEngine: Proactive Learning with Lifelong Memory via World Models*. Empirical Methods in Natural Language Processing (EMNLP, Main Conference), 2026. **Oral (Top 1%)**

Huanhuan Xiao, **Xiaoying Liao**, Céline Portal, Zhenzhen Pan, Eileen Chen, Igor A. Butovich, Carlo Iomini. *Hedgehog Signaling Maintains Meibomian Gland Progenitors and Lipid Homeostasis with Implications for Meibomian Gland Dysfunction Therapeutics*. Investigative Ophthalmology & Visual Science, 2026. pp. 3883.

Nianping Liu, Xuanzhi Chen, Miao Cui, **Xiaoying Liao**, Xiaoke Song, Sairam Pantham, Weize Xu, Xiaojie Qiu. *Bio-Babel: Autonomous Cross-Language Reconstruction of Agent-Ready Computational Biology Ecosystems*. bioRxiv, 2026.

Tianyang Wang, Xi Xiao, **Xiaoying Liao**. *Foundation Models for Radiology Report Generation*. One chapter in medical field's book, 2026. Under review.

Xi Xiao, Yunbei Zhang, Lin Zhao, Yiyang Liu, **Xiaoying Liao**, Zheda Mai, Xingjian Li, Xiao Wang, Hao Xu, Jihun Hamm, Xue Lin, Min Xu, Qifan Wang, Tianyang Wang, Cheng Han. *Prompt-based Adaptation in Large-scale Vision Models: A Survey*. Transactions on Machine Learning Research, 2026.

Junjie Li, Xi Xiao, Yunbei Zhang, Chen Liu, Lin Zhao, **Xiaoying Liao**, Yingrui Ji, Janet Wang, Jianyang Gu, Yingqiang Ge, Weijie Xu, Xi Fang, Xiang Xu, Tianchen Zhao, Youngeun Kim, Tianyang Wang, Jihun Hamm, Smita Krishnaswamy, Jun Huan, Chandan K. Reddy. *Agent Harness Engineering: A Survey*. Under review at ACL, 2026.

Tianyang Wang, Xi Xiao, Gaofei Chen, **Xiaoying Liao**, Guo Cheng, Yingrui Ji. *Boosting Active Learning with Knowledge Transfer*. 2025 IEEE 37th International Conference on Tools with Artificial Intelligence, 2025. pp. 1124–1131.

Xiaoying Liao. *Assessing the Performance of Clustering Methods in Spatially Resolved Transcriptomics Data*. University of Sydney, 2023. Undergraduate honors research.

RESEARCH EXPERIENCE

Stanford University

2026–Present

Research Assistant, supervised by Prof. Xiaojie Qiu

Stanford, CA

- Human fetal heart development:** Analyzing PCW7 and 8 human fetal heart spatial transcriptomics data to characterize cardiac cell states, spatial niches, cell–cell communication, and 3D reconstruction during early heart development.
- 3D embryonic development:** Building a 3D spatiotemporal atlas of mouse embryogenesis from E5.5–E7.5 to characterize cell-state transitions, spatial tissue organization, and developmental trajectories during early embryogenesis.
- Pediatric cardiomyopathy:** Leading a single-cell multi-omics and spatial transcriptomics study comparing myocardial states before and after ventricular assist device support (VAD) and between recovery and non-recovery to identify cellular programs, spatial niches, and potential biomarkers associated with myocardial recovery.

Johns Hopkins University

2024–2026

Research Assistant, supervised by Prof. Carlo Iomini

Baltimore, MD

- Meibomian gland development:** Investigated how Hedgehog-responsive epithelial progenitors regulate gland development, meibocyte maturation, and lipid homeostasis.
- Cilia–Hedgehog signaling:** Characterized Ift88–Smo genetic interactions and their effects on epithelial proliferation, gland morphology, and lipid production and composition.
- Single-cell transcriptomics:** Generated and analyzed P5 and P10 meibomian gland scRNA-seq data to characterize developmental cell states and pathways associated with gland maturation and Hedgehog signaling.

- **Data preprocessing:** Used Python and R to preprocess spatial transcriptomics datasets, including quality control, normalization, feature selection, and dimensionality reduction using principal component analysis (PCA).
- **Spatial domain detection:** Applied and compared clustering approaches, including K-means and Seurat, alongside spatially informed methods such as BayesSpace and STitch3D, to identify tissue domains and characterize spatial organization.
- **Method evaluation:** Benchmarked clustering performance across simulated and real datasets, examining factors affecting method efficacy and identifying strengths and limitations to inform method selection in spatial transcriptomics research.

TEACHING EXPERIENCE

- **Cell & Tissue Engineering Lab:** Mentored students in practical modules on cell culture, gene delivery, metabolic glycoengineering, and tissue modeling. Provided support in data collection, data analysis, and scientific lab reports.

- **STAT2011 Probability and Estimation Theory:** Guided students through probability theories and statistical distributions, including practical R programming for data analysis applications.
- **DATA2002 Data Analytics: Learning from Data:** Led workshops on data analytics techniques, enabling students to apply statistical programming to real-world data challenges and machine learning concepts for informed decision-making.

PROFESSIONAL EXPERIENCE

- Analyzed large-scale retail and consumer datasets using SQL, Python, and R, applying time-series forecasting, regression, and statistical modeling.
- Developed automated analytics pipelines and dashboards to support inventory planning, consumer behavior analysis, and operational decision-making.

- Analyzed global business and user behavior data using SQL and Python, including automated metric monitoring and interactive BI dashboards.
- Applied statistical analysis and user segmentation models to characterize consumer behavior and support data-driven product and marketing strategies.

ACADEMIC ACTIVITIES

Stanford Genetics Department Retreat: Poster presentation	2026
Association for Research in Vision and Ophthalmology (ARVO): Poster presentation	2026
Annual Wilmer Research Meeting: Poster presentation	2025
Sydney Bioinformatics Research Symposium (SBRS): Participant	2023
Unlocking Single Cell Spatial Omics Analyses with scDney: Participant	2023

SKILLS & HONORS

Computational: Python, R, SQL, MATLAB, PyTorch, scikit-learn, Scanpy, Seurat; machine learning, statistical modeling, single-cell and spatial transcriptomics, multi-omics integration

Experimental: 10x Genomics single-cell sequencing, Illumina spatial transcriptomics, cell and tissue culture, immunostaining, Confocal and two-photon microscopy, PCR/qRT-PCR, Western blotting, DNA/RNA extraction

Honors: Vice Chancellor’s Global Mobility Scholarship (2020); Dalyell Scholar (2022); 2nd Prize, China National High School Biology Olympiad (2018)